

```

8
9  x = np.arange(0,20,2)
10
11  plt.plot(x, [xi**2 for xi in x])
12
13  plt.show()

```

We can also add labels, legends, the grid and axis name in the plot. Take a look at Figure 6, and the following code:

```

1  import matplotlib.pyplot as plt
2  import numpy as np
3
4  x = np.arange(0,20,2)
5
6  plt.title('Sample Plot')
7
8  plt.xlabel('X axis')
9
10 plt.ylabel('Y axis')
11
12 plt.plot(x, [xi**3 for xi in x], label='Fast')
13
14 plt.plot(x, [xi**4 for xi in x], label='Slow')
15
16 plt.legend()
17
18 plt.grid(True)
19
20 plt.show()
21
22 plt.savefig('plot.png')

```

You can create various types of plots using Matplotlib. Let us take a look at Pie Plot and Scatter Plot.

```

1  import matplotlib.pyplot as plt
2
3  plt.figure(figsize=(10,10));
4
5  plt.title('Distribution of Dark Energy and Dark Matter in
the Universe')
6
7  x = [74.0,22.0,3.6,0.4]
8
9  labels = ['Dark Energy', 'Dark Matter', 'Intergalactic gas',
'Stars,etc']
10
11 plt.pie(x, labels=labels, autopct='%1.1f%%');
12
13 plt.show()
14
15 import matplotlib.pyplot as plt

```

```

2
3  import numpy as np
4
5  plt.title('Scatter Plot')
6
7  x = np.random.randn(200)
8
9  y = np.random.randn(200)
10
11 plt.xlabel('X axis')
12
13 plt.ylabel('Y axis')
14
15 plt.scatter(x,y)
16
17 plt.show()

```

Similarly, you can plot Histograms and Bar charts using the *plt.hist()* and *plt.bar()* functions, respectively. In our next example, we will generate a plot by using data from a text file:

```

1  import matplotlib.pyplot as plt
2  import numpy as np
3
4  data = np.loadtxt('ndata.txt')
5  x = data[:,0]
6  y = data[:,1]
7
8  figure(1,figsize=(6,4))
9
10 grid(True)
11 hold(True)
12 lw=1
13
14 xlabel('x')
15
16 plot(x,y,'b',linewidth=lw)
17
18 plt.show()

```

After executing this program, it results in the plot shown in Figure 10. So, what's happening here? First of all, we fetch data from the text file using the *loadtxt* function, which splits each non-empty line into a sequence of strings. Empty or commented lines are just skipped. The fetched data is then distributed in variables using *slice*. The *figure* function creates a new figure of the specified dimensions, whereas the *plot* function creates a new line plot.

Mathematical modelling

Now that we have a basic understanding of various computation tools, we can move on to some more complex